MITK User Guide - PA2 Deep Learning

Medical Imaging Interaction Toolkit (MITK Workbench)

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1. Introduction

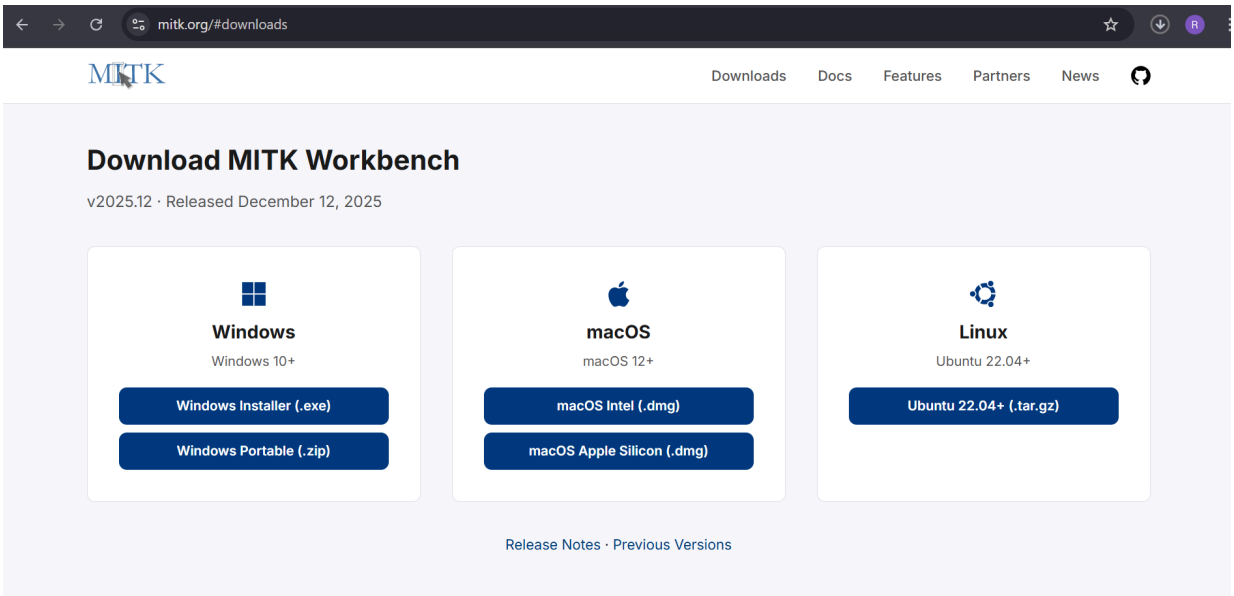
The Medical Imaging Interaction Toolkit (MITK) is an open-source software platform used for medical image analysis, visualization, and segmentation. It supports DICOM, NIfTI, NRRD and other medical imaging formats and is widely used in research and clinical prototyping.

This manual explains how to download, install and work with MITK on **Windows and macOS**.

2. Open the Official Download Page

Go to the official MITK page:

👉 https://www.mitk.org/#downloads



Alternate Link to download: Go to the MITK SourceForge files page:

👉 Open MITK SourceForge Download Page

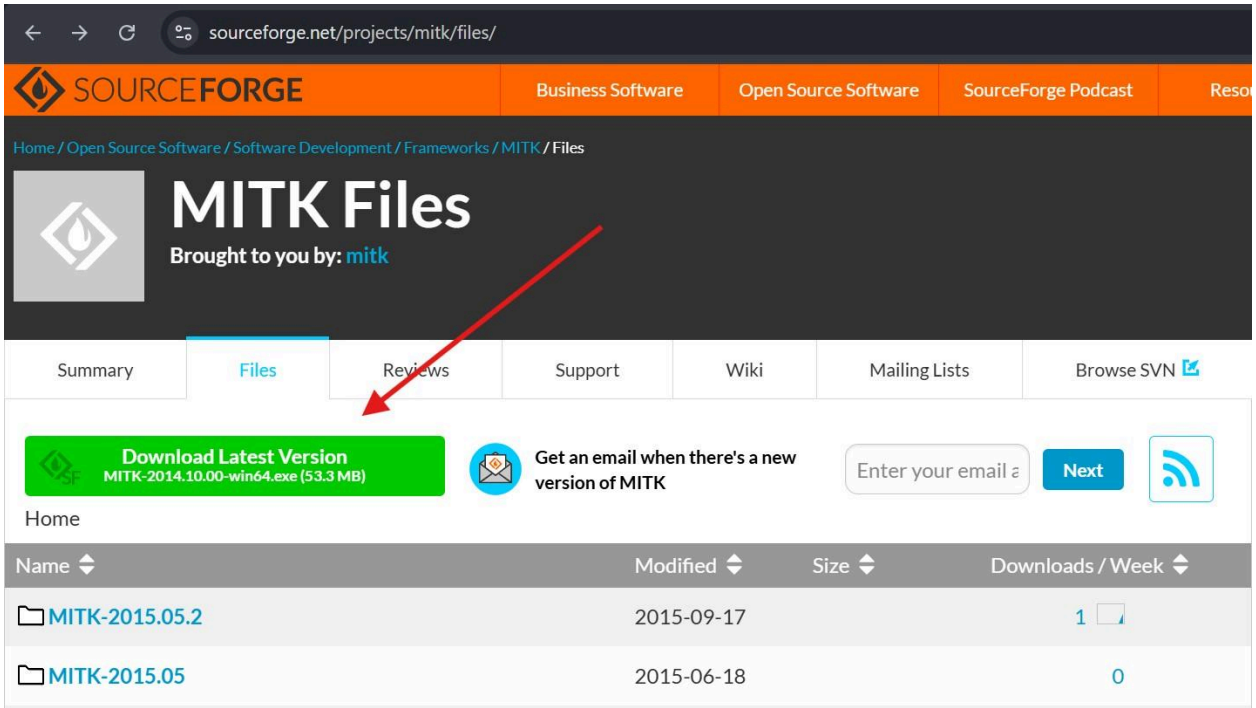
This page contains all available MITK releases.

4. Download MITK (Step-by-Step)

Step 1 — Select relevant option and download from official link (image attached above).
1.1 Alternate Link: SourceForge Page

On the page, click the newest folder available:

(reference image attached below)



For Mac, it will show relevant Mac versions. Click on the green icon (Download Latest version).

5. Download for Windows

Download the file named similar to:

MITK-Workbench-Windows-x86_64.zip

6. Download for macOS

Download the file named similar to:

MITK-Workbench-MacOS.dmg

7. Installation on Windows

Step 1 — Extract the ZIP file

After download:

1. Right-click the ZIP file
2. Click **Extract All**
3. Extract to:

C:\MITK\

Step 2 — Run MITK Workbench

Open the extracted folder and double-click:

MitkWorkbench.exe

8. Installation on macOS

Step 1 — Open the DMG file

Double-click the downloaded .dmg file.

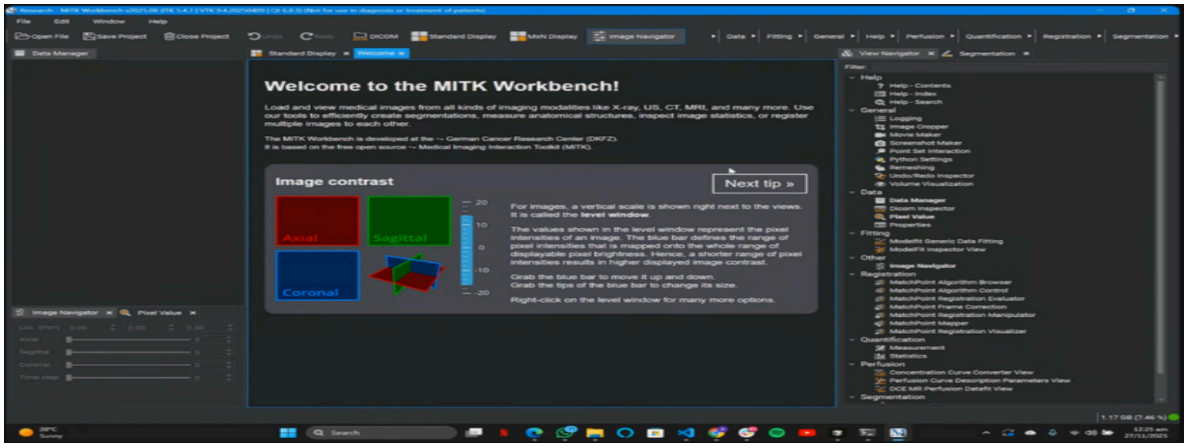
Step 2 — Move to Applications

Drag MITK Workbench into the Applications folder.

The Desktop shortcut icon should look like this:

Step 3 — First Launch

1. Go to Applications
2. Right-click MITK Workbench or use desktop shortcut
3. Click Open



9. Opening an Image in MITK Workbench

After installing MITK, the first task is to load and visualize medical images.
MITK can open:

- DICOM (.dcm)
- NIfTI (.nii / .nii.gz)
- NRRD (.nrrd)
- PNG/JPG/TIFF (2D images)

For this course, we mainly use NIfTI (.nii) files.

Step 1 — Open an Image File

1. Click File on the top menu
2. Click Open File(s)

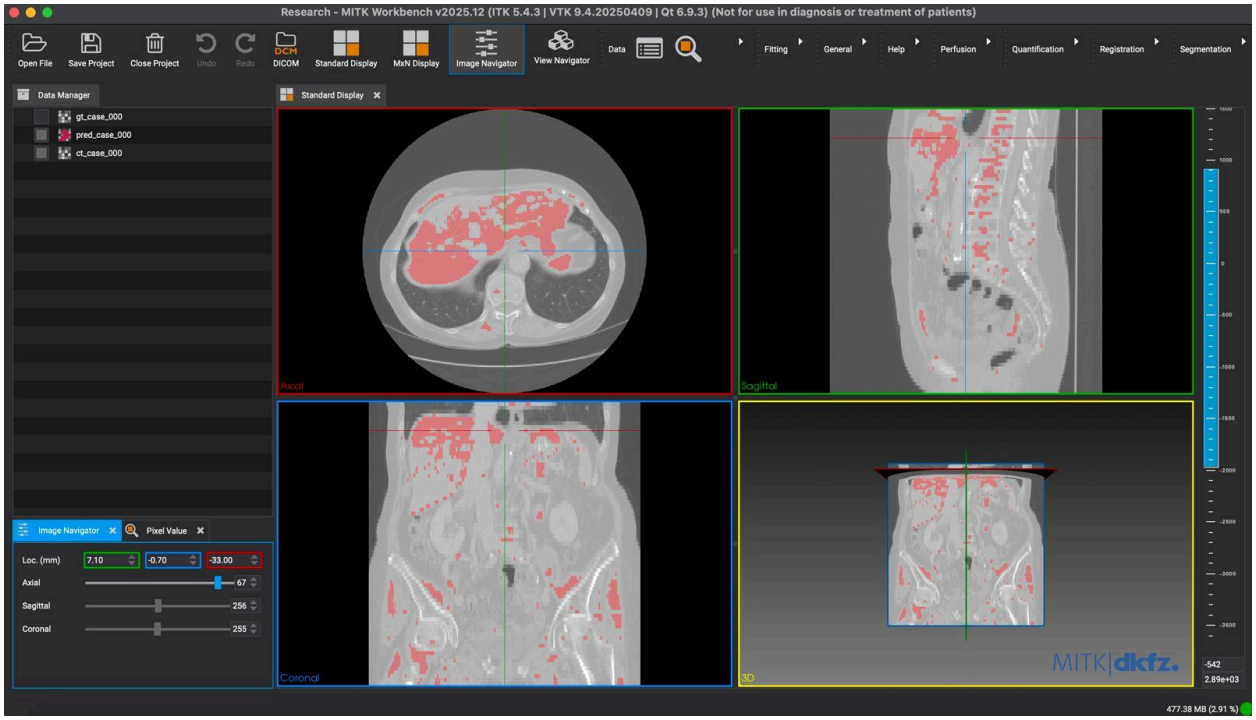
Step 2 — Select the Image

Navigate to the folder containing your results.

For example (LiTS dataset):

Select the file and click Open. You will select the ground truth e.g "gt_case_000" , prediction file e.g "pred_case_000" and ct file e.g"ct_case_000" simultaneously and open them in MITK.

MITK will load the image automatically. (reference image attached below)



Step 3 — Verify the Image Loaded

After loading, MITK shows the image in four views:

- Axial view (top view)
- Sagittal view (side view)
- Coronal view (front view)
- 3D view

This means the 3D medical image loaded successfully ✅